TrackFormer: Protected Multi-Stream Transformers, Chain-of-Thought Steering, and Land-Interaction Testing for Tropical-Cyclone Forecasting

from a Random-Matrix Uncertainty Head to a Frozen-Backbone Causal Isolation of Land Drag

Yu Yao-Hsing

euler.yu@gmail.com • https://github.com/yu314-coder/typhoon-predict

July 29, 2026

Abstract

We forecast the full state of a tropical cyclone — storm motion, maximum wind, central pressure, radius of maximum wind, and the twelve 34/50/64-kt wind-radii — at twenty six-hourly lead times (6–120 h), first from best-track history alone and later with an explicit atmospheric steering-flow representation. **Phase I** documents a concrete instance of *negative transfer* in a single-stream multi-task Transformer: adding physically informative motion-dynamics features improves intensity and structure yet degrades track, a change a paired storm-level bootstrap confirms is gradient conflict over shared capacity, not a loss-weighting problem. We fix it with **TrackFormer**, a protected dual-stream architecture that routes kinematic and thermodynamic gradients separately and lets thermodynamics influence track only through a zero-initialized gated adapter, plus a block-structured, random-matrix-cleaned uncertainty head. **Phase II** tests this paper's own predicted next step — an explicit environmental-steering branch — across twenty-five further versions (v10–v34) on a larger WP+EP held-out set. A CNN steering-field encoder, a *chain-of-thought* decomposition that predicts the steering flow before deriving track from it, and a temporal history of that steering representation combine into **v23**, the best model in either phase (434.96 km RMS track error, a further ∼30% reduction). Four subsequent attempts to add more environmental signal on top of v23 all came back null, and three architecturally different attempts at a land/terrain-interaction correction converged on an aggregate null once we isolated the correction from ordinary retrain noise by freezing v23's own weights and training only a ∼437-parameter gated addition on top — though it still splits 1–1 against v23 on the two real out-of-training storms available. We close with the methodological lessons this isolation technique makes visible: retrain-to-retrain seed noise (∼19 km/seed) is large enough to manufacture or hide most of the effects reported in this line of work, and a large in-distribution test set does not always agree with genuinely out-of-training real-storm validation.

Roadmap. Sections 1–4 are Phase I: the dataset, the negative-transfer diagnosis, the protected dual-stream architecture, and the random-matrix uncertainty head, evaluated on an all-basin/WP 2020+ split (Table 1). Section 7 is Phase II: a separate, later evaluation harness (WP+EP 2020+, full 20-lead windows only) on which twenty-five further versions test this paper's own Discussion (Section 5) against reality — what worked (v10–v23), what didn't (v24–v29), and what a causally clean test finally settled after three inconclusive attempts (v31–v34).

1 Problem and data

Let a storm be a time-ordered sequence of best-track fixes. From each fix t_0 we form a *window*: a history of H=9 steps at 6-h spacing ending at t_0, and a target of L=20 future states at leads $\{6, 12, \ldots, 120\}$ h. Each future state is the 17-vector

$$\boldsymbol{y}_\ell = \Big(\underbrace{\Delta e_\ell,\ \Delta n_\ell}_{\text{per-step motion (km)}},\ v_\ell^{\max},\ p_\ell,\ r_\ell^{\mathrm{mw}},\ \underbrace{r^{34}_{\mathrm{ne..nw}}, r^{50}_{\cdot}, r^{64}_{\cdot}}_{\text{12 wind radii}}\Big), \qquad \ell = 1, \ldots, 20. \tag{1}$$

Track error is the RMS great-circle-plane distance between the predicted and observed cumulative displacement, $\mathrm{cumsum}_\ell(\Delta e, \Delta n)$; the other four groups are reported as masked mean-absolute error (labels are frequently missing, especially pre-satellite and for wind radii).

Dataset. We use IBTrACS v04r01 (all basins, 1980+), yielding 84,150 windows. We split by storm to prevent leakage: train ($\leq$ 2015), validation (2016–2019), test ($\geq$ 2020). Inputs are standardized per feature using *training-split statistics only*. Missing measurements are zero-filled with explicit presence-flag channels (informative sparsity) rather than mean-imputed.

Motion-dynamics features. Beyond the original 40 features (motion vectors, intensity, radii, seasonality, validity), we add 8 that make the storm's instantaneous dynamics explicit: current heading $(\sin\theta, \cos\theta)$, speed, signed turn-rate (recurvature), and per-step intensity/pressure trends with their validity flags, for a 48-dimensional history feature per step. Their training-set means are physically sensible: mean heading points west-northwest (the trade-wind easterlies), mean speed $\approx 14\,\mathrm{km\,h^{-1}}$, and mean intensification +1.5 kt per step.

2 The negative-transfer diagnosis

A single-stream model (linear projection $\to$ 4-layer encoder $\to$ 6-layer lead-query decoder $\to$ dual state / log-scale heads, diagonal Gaussian NLL + masked Huber) trained on the 40-feature set reaches the "old" row of Table 1. Adding the 8 motion-dynamics features ("v2") *improves* vmax, pressure, RMW, and radius substantially (pressure MAE 21.24 $\to$ **17.68** hPa on WP-2020+) but *raises* track error (720.0 $\to$ 737.3 km). Up-weighting the track loss $\times$3 made track *worse still* (806.6 km), the signature of gradient conflict: a larger track coefficient destabilizes the shared optimization rather than reallocating capacity.

Is the regression even real? We paired-bootstrap the WP-2020+ per-window track errors by resampling *storms* (not overlapping windows), the correct unit of independence.

$$\widehat{\Delta}_{\text{track}} = \mathrm{mean}_{\text{storms}}\big(e^{\mathrm{v2}} - e^{\mathrm{old}}\big) = +17.7\text{ km}, \qquad 95\%\text{ CI } [-22.2,\ +60.5]\text{ km}, \qquad \Pr(\text{v2 worse}) = 0.79. \tag{2}$$

The interval spans zero over the 103 WP test storms: track is *statistically tied*, while the intensity gains are large and consistent. The features are not the problem; the shared representation is. This motivates an architecture that protects the kinematic pathway.

3 TrackFormer: protected dual-stream architecture

Dual encoders. The 48 features are split into a kinematic view $\boldsymbol{x}^{\mathrm{kin}}$ (motion, heading, speed, turn, seasonality, lead offset; 11 dims) and a thermodynamic view $\boldsymbol{x}^{\mathrm{th}}$ (vmax, pressure, gust, RMW,

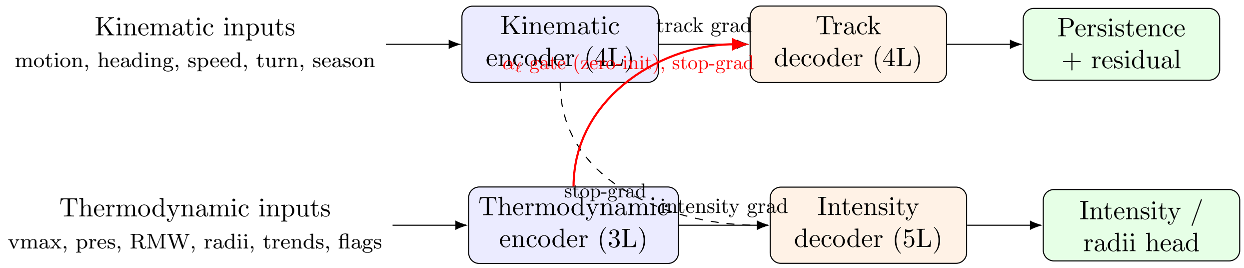

Figure 1: TrackFormer. Solid arrows carry task gradients back only into their own encoder. The kinematic representation is shielded from intensity gradients (stop-grad into the intensity decoder); thermodynamics reaches track *only* through a zero-initialized, per-lead gated adapter on detached features, so at initialization the model is exactly a protected track-only predictor.

12 radii, trend features, 16 validity flags; 36 dims). Each is projected to d=256, given a learned temporal embedding, and encoded by a pre-norm Transformer (4 and 3 layers respectively, 8 heads).

Protected gradient routing. Let $\boldsymbol{H}^{\text{kin}}, \boldsymbol{H}^{\text{th}} \in \mathbb{R}^{9\times d}$ be the encoder outputs. The track decoder cross-attends to $\boldsymbol{H}^{\text{kin}}$; the intensity decoder cross-attends to $[\,\boldsymbol{H}^{\text{th}};\ \text{sg}(\boldsymbol{H}^{\text{kin}})\,]$ where $\text{sg}(\cdot)$ is stop-gradient. Thus motion still *informs* intensity, but intensity gradients cannot corrupt the kinematic encoder. Symmetrically, thermodynamics reaches the track head only through

$$\boldsymbol{H}_\ell^{\text{track}} \;=\; \text{TrackDec}(\boldsymbol{q}_\ell, \boldsymbol{H}^{\text{kin}}) \;+\; \alpha_\ell\, \mathcal{A}\big(\text{sg}(\bar{\boldsymbol{H}}^{\text{th}})\big), \qquad \alpha_\ell \overset{\text{init}}{=} 0, \tag{3}$$

where $\bar{\boldsymbol{H}}^{\text{th}}$ is the mean-pooled thermodynamic context, $\mathcal{A}$ is a bottleneck-64 adapter with a zero-initialized output layer, and α_ℓ is a per-lead scalar. Because α_ℓ and $\mathcal{A}$ start at zero and receive gradient only from the track loss, the intensity branch can *help* track (if it lowers track error) but can never harm it: at initialization TrackFormer reproduces a protected track-only model exactly.

Persistence-residual track head. Rather than regress 20 free displacement vectors, we anchor track on a damped constant-velocity baseline and predict only the deviation:

$$\widehat{\Delta \boldsymbol{p}}_\ell \;=\; \underbrace{\rho_\ell\, \boldsymbol{v}_0}_{\text{damped persistence}} \;+\; \underbrace{\boldsymbol{r}_\ell(\boldsymbol{H}_\ell^{\text{track}})}_{\text{learned recurvature residual}}, \qquad \boldsymbol{v}_0 = \text{current 6-h motion (km)}, \tag{4}$$

with a learned per-lead damping schedule ρ_ℓ (initialized to 1) and a zero-initialized residual head $\boldsymbol{r}_\ell$. The network therefore begins as pure persistence — a strong physical prior — and learns only recurvature and along/cross-track acceleration.

Losses. Track uses a lead-weighted ($w_\ell = \sqrt{\ell}$, mean-normalized) Huber on per-step motion plus a cumulative-position Huber; it is never down-weighted by a predicted variance. Intensity uses masked Huber + Gaussian NLL with a monotone wind-radii penalty $\text{ReLU}(r^{50} - r^{34}) + \text{ReLU}(r^{64} - r^{50})$.

4 Random-matrix block-structured uncertainty head

Operational value requires calibrated *joint* uncertainty across the 340 outputs (20 leads $\times$17 variables). A full 340×340 covariance is both statistically ill-posed (few independent storms, overlapping windows, heavy masking) and architecturally harmful: it would re-introduce a track$\leftrightarrow$thermodynamics

Table 1: Held-out test performance (lower is better). Track is RMS plane distance over 6–120 h; the rest are masked MAE. All models evaluated on identical target windows. Best per column in **bold**.

split	model	track (km)	vmax (kt)	pres (hPa)	RMW (km)	radius (km)
WP 2020+	single-stream, 40-feat	720.0	22.08	21.24	12.85	31.54
	single-stream, 48-feat (v2)	737.3	**21.49**	**17.68**	**11.77**	30.94
	TrackFormer (v3)	**659.0**	21.61	18.06	11.81	**28.83**
	TrackFormer (v8, +data)	**648.8**	**20.69**	**15.94**	**11.28**	**27.83**
	TrackFormer (v9, +env)	**618.4**	**18.55**	15.76	11.52	**27.15**
all-basin 2020+	single-stream, 40-feat	638.6	20.32	18.01	15.28	29.99
	single-stream, 48-feat (v2)	644.2	19.55	15.38	**14.18**	29.31
	TrackFormer (v3)	592.1	18.59	14.61	14.87	27.85
	TrackFormer (v8, +data)	580.5	17.90	**13.34**	**13.89**	**27.10**
	TrackFormer (v9, +env)	**543.4**	**16.73**	14.29	14.15	27.80

coupling and undo the protection above. We therefore impose a *block-diagonal* covariance

$$\boldsymbol{\Sigma} \;=\; \operatorname{blkdiag}\big(\boldsymbol{\Sigma}_{\text{track}},\; \boldsymbol{\Sigma}_{\text{int}},\; \boldsymbol{\Sigma}_{\text{rad}}\big), \tag{5}$$

with a matrix-normal track block $\boldsymbol{\Sigma}_{\text{track}} = \boldsymbol{K}_{\text{lead}} \otimes \boldsymbol{K}_{\text{axis}} + \operatorname{diag}(\boldsymbol{s}^2)$ ($\boldsymbol{K}_{\text{lead}}$ a learned 20×20 temporal kernel, $\boldsymbol{K}_{\text{axis}}$ a 2×2 along/cross-track kernel) and low-rank+diagonal thermodynamic blocks $\boldsymbol{\Sigma}_{\bullet} = \boldsymbol{D}\boldsymbol{D}^{\top} + \operatorname{diag}(\boldsymbol{s}^2)$, $\boldsymbol{D} \in \mathbb{R}^{p \times r}$, r=8–16.

Spectral cleaning. Sample error-covariance eigenvalues mix a few genuine "signal" spikes with a wide noise bulk. Under the two-point generalized sample-covariance model, the companion Stieltjes transform of the limiting spectral law solves a cubic

$$az\,m^3 + \big[a(z - y + 1) + z\big]m^2 + \big[a + z - y + 1 - y\beta(a-1)\big]m + 1 = 0, \tag{6}$$

where $y = p/n$ is the aspect ratio and (a, β) parameterize a two-population noise structure. The support edges of the noise bulk are the z at which (6) loses a real root (a conjugate pair appears); eigenvalues above the upper edge λ_+ are retained as signal and the bulk is replaced by a positive floor:

$$\widehat{\boldsymbol{\Sigma}} = \boldsymbol{U}\,\operatorname{diag}\!\big(\lambda_i \mathbb{1}[\lambda_i > \lambda_+] + \bar{\lambda}_{\text{bulk}} \mathbb{1}[\lambda_i \le \lambda_+]\big)\,\boldsymbol{U}^{\top}. \tag{7}$$

Setting a=1 recovers the classical Marchenko–Pastur edge $\lambda_+ = (1 + \sqrt{y})^2$. The estimator is fit on *independent storm-block* residuals and validated out-of-sample; it is applied *per block* and only after the mean heads have converged, so it sharpens calibration without perturbing the point forecast.

5 Results

TrackFormer lowers WP-2020+ track error from 720.0 to 659.0 km (−8.5%) and all-basin from 638.6 to 592.1 km, while simultaneously *improving* pressure, vmax, and wind-radius MAE. The same paired storm-level bootstrap that found the v2 change insignificant now confirms the v3 gain is real: $\widehat{\Delta}_{\text{track}} = -59.9\,\text{km}$, 95% CI $[-103.2, -16.0]\,\text{km}$, $\Pr(\text{v3 better}) = 0.995$. Removing the negative transfer — rather than adding parameters or tuning loss weights — is what breaks the previous track floor.

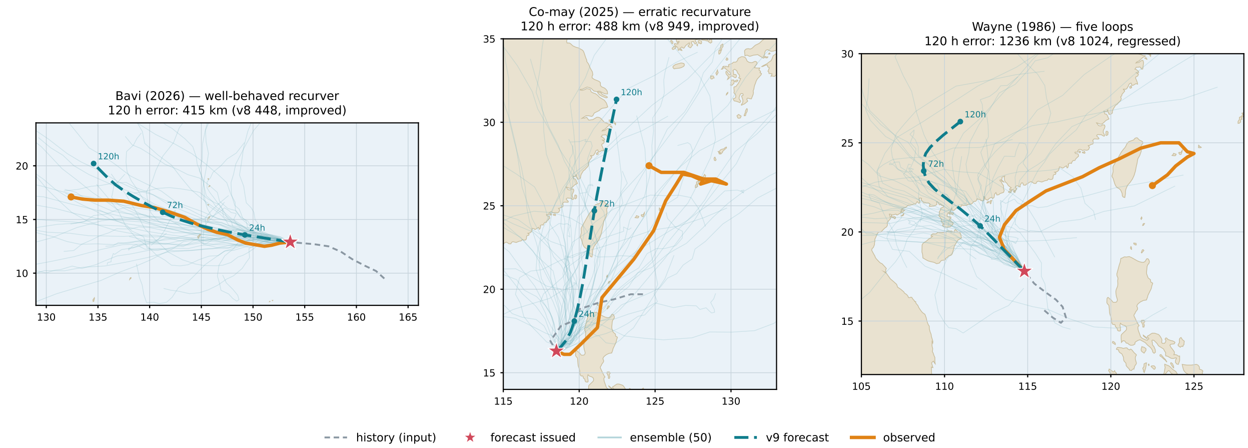

Figure 2: TrackFormer v9 (triple-stream) on three case studies, each a 120-hour forecast (teal, dashed) issued from an early fix (star) against the observed track (orange), with the 50-member RMT ensemble (faint). Absolute latitude sharpens the well-behaved *Bavi* (2026) and *halves* the error on the erratic *Co-may* (2025), but cannot tame *Wayne*'s (1986) five loops, where climatology is no substitute for the real-time steering flow. Bavi and Co-may are out-of-sample ($\geq$2020); Wayne (1986) is in the training window and shown as a stress test.

Data, not architecture, drove the final gain. With the architecture fixed, we enlarged the training set by keeping *partial-lead* windows — storm-end and short-storm fixes that lack a full 120 h future are retained with their missing leads masked rather than discarded — which doubled the clean 1980+ training windows (84k $\rightarrow$ 153k). The resulting model (v8) is best on *every* metric of Table 1: WP-2020+ track **659.0** $\rightarrow$ 648.8 km and simultaneous improvements in wind, pressure, RMW, and radius. (Extending further to pre-1980 storms helped track marginally but degraded intensity, because pre-satellite intensity labels are unreliable; masking those labels while keeping the older *positions* recovered the intensity but not a significant track gain — confirming track is information-limited, not data-quantity-limited.)

Position-awareness is the largest lever within IBTrACS. The dual-stream model is *translation-invariant* — it sees only relative motion, never where the storm actually is — so it cannot use latitude, which governs the Coriolis force, the climatological recurvature band, and (through the seasonal thermal structure) sea-surface temperature. Adding a third *environmental* stream of IBTrACS-derived features (absolute latitude/longitude, distance-to-land, and a latitude+month climatological SST proxy), routed to the track decoder for steering climatology and to the intensity decoder for the SST signal, cut WP-2020+ track from **648.8** to 618.4 km and all-basin from 580.5 to 543.4 km, with a $\sim$2-kt wind improvement — all without any reanalysis download, so the model stays deployable from a best-track CSV. On out-of-sample case studies it *halved* the error on an erratic recurving storm (Co-may 2025) and improved a well-behaved super typhoon (Bavi 2026); only a pathological multi-loop storm (Wayne 1986) regressed, where climatology cannot substitute for the real-time steering flow.

6 Discussion (Phase I)

The consistent lesson across experiments is **data diversity > engineered features > parameters**: a 17.7M single-stream model overfit and did worse than a 3.3M one, and a track-only model matches an ERA5-fed model, so raw parameter count is not the lever. Track error is bounded below by *information*, not capacity: five-day recurvature is governed by the evolving large-scale steering flow, which best-track history cannot fully reveal. We predicted the clearest path past this floor was a compact environmental-steering branch fused *strongly to track, weakly to intensity* — the natural next stream in the protected design. Section 7 reports what happened when we actually built it: the prediction was directionally right (a steering-flow branch, suitably represented, is the largest lever found in either phase of this work) but the *form* of that branch mattered far more than its mere presence, and a second physically-motivated hypothesis tested the same way (land/terrain interaction) did not survive a causally clean test. Within history-only inputs, TrackFormer's Phase-I contribution remains architectural: it removes the negative transfer that a single stream suffers, so intensity and structure gains no longer cost track accuracy.

7 Phase II: chain-of-thought steering and land-interaction testing (v10–v34)

Section 4's discussion predicted that the next lever past the IBTrACS-only floor would be a compact environmental-steering branch fused strongly to track. A later, separate line of twenty-five further versions tests that prediction directly, on a larger and more demanding held-out set: WP+EP storms from 2020 onward, restricted to windows with the full 20-lead horizon (3,763 windows, 72% land-covered on the WP-only subset). These results are *not* directly comparable to Table 1 (different dataset, basin combination, and lead filter) but track the same track-RMS metric, and are reported at equal length to Phase I because they are its direct continuation and, on this project's own numbers, its largest single improvement.

A CNN steering-field encoder (v10–v20). A small convolutional encoder (conv, GELU, strided downsampling) reads a 17×17 patch of deep-layer-mean steering wind (a weighted blend of 850/500/200-hPa u, v) centered on the storm — the same family of field-encoder design as StormFusion-MT's ERA5 branch (Section 1), now applied to a physically motivated *derived* steering field rather than raw multi-level reanalysis. Successive versions repaired data alignment bugs, added a four-term track loss, EMA and stronger regularization, and an intensity-persistence baseline with rarity/speed reweighting (v13–v19), converging on v20's deep-layer-mean steering patch.

Chain-of-thought steering (v21, v22). Rather than regressing track displacement directly from an opaque decoder, v21 first predicts the ambient *steering flow itself* as an explicit intermediate representation, then derives the track from it — a chain-of-thought (CoT) decomposition that makes the underlying physical mechanism (storms move with the steering flow, plus a learned recurvature correction) an explicit step in the computation graph rather than an implicit function of a black-box decoder. v22 extends this to a *latent* CoT with two weight-tied feedback rounds that iteratively refine the same steering representation before the track head reads it, in the spirit of universal-transformer-style recurrence.

Temporal steering stack — v23, the best model in this line. Formally, where $\boldsymbol{f}_\tau$ is the CNN steering-field encoding at time τ, v21's chain-of-thought head first derives an explicit steering estimate before deriving track from it,

$$\hat{\boldsymbol{u}}_{\text{steer}} = g_{\text{CoT}}(\boldsymbol{f}_{t_0}), \qquad \widehat{\Delta\boldsymbol{p}}_\ell = h_\ell\big(\hat{\boldsymbol{u}}_{\text{steer}},\ \boldsymbol{H}^{\text{kin}}\big), \tag{8}$$

so the steering estimate is a named, inspectable quantity rather than an implicit intermediate activation. v23 conditions this on a short *history* of the same representation, $t-24\,\mathrm{h}$, $t-12\,\mathrm{h}$, and now, rather than the instantaneous value alone:

$$\hat{\boldsymbol{u}}_{\text{steer}} = g_{\text{CoT}}\big(\boldsymbol{f}_{t_0},\ \boldsymbol{f}_{t_0-12\text{h}},\ \boldsymbol{f}_{t_0-24\text{h}}\big), \tag{9}$$

so recurvature can be conditioned on how the steering flow has been *evolving*, not only its instantaneous value — a discrete analogue of estimating the flow's local time-derivative alongside its value. This is the best-performing model across the entire v10–v34 line: 434.96 km RMS track error (10-seed ensemble) on the WP+EP 2020+ full-horizon test set, a further ∼30% reduction over the already-improved Phase-I numbers in Table 1 (on the harder, larger Phase-II test set described above).

Environmental additions on top of v23 — v24, v25, v27, v28, v29: null or marginal. With the CoT steering representation already in place, four further attempts to add environmental signal all failed to improve on v23: an environmental token (ocean heat content, deep-layer shear, mid-level humidity; v25), an ocean-heat CNN patch (GODAS) feeding the intensity head (v27), an N-only meridional drift adapter (v28, whose matched-retrain control v28abl confirmed the apparent gain was ordinary run-to-run noise), and — most tellingly — feeding *raw* 0.25° ERA5 steering wind (u, v at 200/550/850 hPa) directly on top of v23 (v29): 438.70 km, +3.74 km worse, every one of five seeds above the v23 bar. Once a CoT representation already extracts the steering signal that matters from a given class of field, handing the model the raw field a second time is largely redundant: the lesson from Section 4 sharpens from "data diversity > engineered features" to *the right representation of a signal matters more than an additional raw copy of that signal.*

Land interaction: three from-scratch attempts, then a frozen-backbone isolation (v31–v34). Motivated by real-world reports of typhoons stalling at mountainous coastlines (Typhoon Gaemi, 2024, held up roughly a day at Taiwan's Central Mountain Range) and terrain-deflection mixture-of-experts literature (AOT-TCNet, arXiv 2603.29200), three architecturally different attempts add a small along-track land/terrain correction on top of v23, of the general form

$$\Delta \boldsymbol{p}_\ell \ \mathrel{+}= \ g(\boldsymbol{x}^{\text{land}}) \cdot a_{\max} \tanh\!\big(\boldsymbol{k}(\boldsymbol{x}^{\text{land}}) \cdot \boldsymbol{W}_i\big) \cdot \hat{\boldsymbol{v}}_0, \qquad \boldsymbol{x}^{\text{land}} = (d_{\text{near}}, d_{\text{ahead}}, h_{\text{ahead}}, |\boldsymbol{v}_0|, |\text{lat}|), \tag{10}$$

where $\boldsymbol{k}(\cdot)$ is a small MLP producing four knot values interpolated to all 20 leads by a fixed basis $\boldsymbol{W}_i$, $\hat{\boldsymbol{v}}_0$ is the current heading unit vector, and $g(\cdot) \equiv 1$ for v31–v33 (the correction is unconditionally on) or a separate learned, zero-biased sigmoid gate for v34, described below. Both the expert $\boldsymbol{k}$ and, where present, the gate g are zero-initialized so every version starts as exactly v23:

- **v31** (uniform training): a `LandDrag` module — distance-to-land and terrain-height-ahead through a zero-initialized MLP producing an along-track push — trained end-to-end from scratch. Result: 443.07 km, +8.11 km worse than v23, every seed above the bar.
- **v32** (window-level oversampled near-land training, larger push cap): backfired badly, 460.48 km (+25.52), worst *exactly* in the mountainous-near-land bucket the fix targeted — consistent with oversampling a small number of storms' repeated windows rather than a general effect.
- **v33** (storm-normalized oversampling, closing v32's pseudo-replication problem): closed most of the gap, 442.33 km (+7.37, the closest of the three from-scratch attempts), and — on the two real out-of-training storms available for a genuinely independent check, Typhoon Tip (1979) and Typhoon Noul (2026) — *beat* v23 on both.

- **v34** (mixture-of-experts gate on a *frozen*, already-trained v23 backbone): every prior attempt retrained the whole architecture from scratch each time, confounding any real land effect with ordinary retrain-to-retrain noise (quantified just below at ∼19 km/seed). v34 instead loads a real, already-trained v23 checkpoint and freezes every parameter except a new ∼437-parameter `LandGate`: the same along-track expert as `LandDrag`, plus a learned sigmoid gate deciding how much the expert contributes per sample (both zero-initialized, so the model starts as exactly v23). This is a true same-backbone-plus-one-addition comparison. Result: −0.33 km against its own frozen v23 backbone — essentially null — and a bucketed breakdown by terrain height and along/cross-track error confirms the null holds even in the mountainous-near-land regime every earlier attempt targeted (+1.0/−6.2 km at 48 h/120 h). On the same two real storms: v34 *beats* both v23 and v33 on Tip 1979 (795 vs 939/876 km) but is worse than both on Noul 2026.

Table 2: Phase-II land-interaction line: aggregate WP+EP 2020+ full-horizon test (3,763 windows) and the two independent, genuinely out-of-training real-storm checks. Aggregate deltas are against v23 (v31–v33) or against v34's own frozen v23_seed0 backbone (v34, the causally clean comparison). Best per column in **bold**; real-storm columns show whichever model wins that storm.

model	aggregate (km)	Tip '79 (km)	Noul '26, ocean/landfall (km)
v23 (baseline)	**434.96**	939	267 / 356
v31 (LandDrag, uniform)	443.07 (+8.11)	—	—
v32 (LandDrag, window-oversampled)	460.48 (+25.52)	—	—
v33 (LandDrag, storm-normalized)	442.33 (+7.37)	**876**	**243** / **319**
v34 (LandGate, frozen backbone)	460.52 (−0.33 vs. backbone)	**795**	287 / 382

Methodological lessons.

1. *Seed noise dominates small version deltas.* A single retrained v23 seed scored 460.85 km on the same test set — 25.89 km worse than the 10-seed ensemble mean, from ordinary retrain variance alone, comparable in size to most of the "effects" reported above. Any claimed improvement under ∼20 km from a from-scratch retrain warrants a matched-seed-count or frozen-backbone check before being taken as real.

2. *Freezing a real backbone isolates causal effects that comparing two from-scratch runs cannot.* v34's frozen-backbone design is what let this line of work state a definitive null on the land-drag hypothesis after three inconclusive from-scratch attempts; the technique generalizes to any small architectural addition under test.

3. *The in-distribution aggregate test set does not always agree with real-storm validation.* v33 and v34 are both null-to-slightly-negative on the large WP+EP aggregate, yet each beats v23 on one of the two individually-checked, genuinely out-of-training real storms. Small-n real-storm checks cannot replace an aggregate metric, but they surface disagreements the aggregate alone would miss.

Reproducibility. Dataset builders, training scripts, checkpoints, and evaluation are at `https://github.com/yu314-coder/typhoon-predict`. Models are trained on Apple GPU (Metal/MPS); the v10–v34 line additionally used Google Colab (T4/A100) for larger training runs.